CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (80 MARKS)

QUESTION 1

a) List two types of malware described in the course and, for each, state one characteristic behaviour that distinguishes it.

(2 marks)

b) State six techniques or behaviours by which malware attacks and propagates through computer systems, as described in the course, covering executable infection methods, historical propagation examples, and how specific malware types operate on infected hosts.

(6 marks)

QUESTION 2

a) Explain how a substitution cipher conceals the original message, and state what serves as the key in such a cipher.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) State the two stages of the biometric authentication process.

(2 marks)

b) Discuss six aspects of biometric mechanisms and technologies used for user authentication, covering how biometrics are defined, how they are categorised, and the characteristics of specific biometric technologies.

(6 marks)

QUESTION 4

a) Explain how the classes of threats described in the course can affect the confidentiality, integrity and availability of information assets.

(2 marks)

b) Refer to your answer in (a). Discuss the Samy worm as a case study of script injection, covering how the malicious script is delivered and executed, how the worm spread itself, and the scale of the resulting infection.

(6 marks)

QUESTION 5

a) Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

(2 marks)

b) Describe six (6) points about the access control matrix, access control lists and capabilities as access control structures, covering what each structure represents and one advantage or disadvantage of each.

(6 marks)

QUESTION 6

a) State two ways in which an intrusion detection system can support the detection and defence against attacks on a computer system.

(2 marks)

b) Describe six threat classes or attack techniques covered in the course and, for each, state one countermeasure or protective mechanism that can be applied against it.

(6 marks)

QUESTION 7

a) Explain how the chain of trust works in the Trusted Computing architecture, referring to the role of the TPM and the measurement of each stage.

(2 marks)

b) State six components and functions of the NGSCB architecture, focusing on the Nexus and the Nexus Computing Agent (NCA).

(6 marks)

QUESTION 8

a) List TWO (2) password security controls that can be applied to password files and user accounts.

(2 marks)

b) Explain how single sign-on, Kerberos and smart cards support user authentication in large distributed environments. Your answer should cover the motivation for single sign-on, the solutions available, how Kerberos handles a user login, and where these mechanisms fit among alternative authentication approaches.

(6 marks)

QUESTION 9

a) Explain the difference between morals, ethics and laws as three distinct forms of guidance on conduct.

(2 marks)

b) Explain six ways in which a professional code of ethics and conduct, such as the ACM code, guides the behaviour of a computing professional.

(6 marks)

QUESTION 10

a) State TWO (2) typical locations where IDS sensors are placed in a network-based IDS architecture.

(2 marks)

b) Discuss the preventive network security mechanisms that an organization can deploy to protect its network, covering firewalls, proxy servers, network segmentation, remote access VPN and intrusion detection/prevention systems. Provide six distinct points.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) Discuss the ethical impacts of technology by examining how professional codes of ethics, such as the ACM code, guide a computing professional's conduct, and evaluate the limitations of such codes in addressing the ethical impacts of technology.

(5 marks)

b) Explain five ethical responsibilities that the hospital's information technology unit owes to patients with respect to the privacy and accuracy of their medical records, drawing on the ethical dimensions of computer security.

(5 marks)

QUESTION 2

a) State FIVE (5) points about software insecurity and buffer overflows, covering their root causes, how a stack buffer overflow hijacks execution, and techniques used to strengthen software security.

(5 marks)

b) Windows provides several security mechanisms for managing users, objects and access. Explain FIVE (5) points about these Windows security mechanisms, covering the directory service used to organise resources, the access rights model, the role of the registry, and why the complexity of these mechanisms can itself be a weakness.

(5 marks)

END OF QUESTION PAPER